

Learning Multivariate Empirical Mode Decomposition for Spectral Motion Editing

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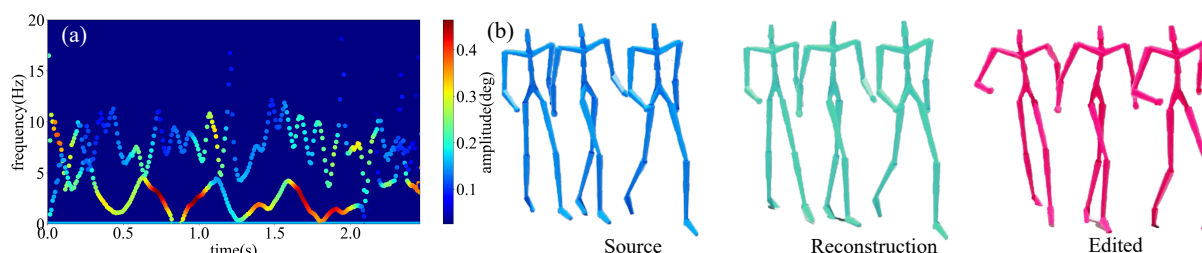


Figure 1: Spectral editing by our proposed decomposition architecture with motion neural networks. (a) Motion features decomposed in the frequency domain by the neural networks (b) The source motion (blue) can be reconstructed (green) and edited (red) by changing the decomposed components shown in the spectrum (a).

ABSTRACT

Spectral motion editing provides an important perspective for character animation synthesis since biomechanical features are usually embedded in the frequency domain. However, heavy manual post-processing and high computational cost were required to achieve high-quality character animations. In this paper, first, we propose a novel architecture for neural networks to learn multivariate empirical mode decomposition that can decompose motion into non-linear frequency components corresponding to their biomechanical features. Next, we demonstrate a spectral motion editing technique based on our proposed architecture. The results revealed that high-quality character animation synthesis could be achieved by editing these decomposed non-linear frequency components, providing novel tasks for character animation design.

CCS CONCEPTS

• **Computing methodologies** → **Motion capture; Motion processing; Neural networks.**

KEYWORDS

Multivariate empirical mode decomposition, Hilbert-Huang transform, Frequency domain, Deep learning, Biomechanics, Motion synthesis, Character animation

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1 INTRODUCTION

Editing motion features in the frequency domain is desirable since the detailed motion features triggered by biomechanical motion hide in the frequency domain, which is important for generating realistic human motion. In the recent studies based on the deep learning method, motion editing in the frequency domain using the neural network has been investigated usually based on the Fourier transform (FT) [Starke et al. 2022], revealing the importance and usefulness of learning motion latent codes in the frequency domain. Nevertheless, although FT is a common method adopted to analyze data in the frequency domain, it has limitations as it non-linearly decomposes signals into trigonometric functions with only mathematical meanings, making it difficult to represent motion features non-linearly. As a result, these deep learning methods using FT suffer to non-linearly edit motion in the frequency domain to generate detailed motion features for achieving high-quality character animations.

On the other hand, the empirical mode decomposition (EMD) introduced by Huang et al. [Huang 2014] is better suited for non-linear data like motion capture data. EMD decomposes non-linear and non-stationary data into pseudo monochromatic waves called the intrinsic mode functions (IMFs) and a residual called Trend. IMFs could be applied to the Hilbert Transform (HT) to analyze motion features in the frequency domain. This process is also known as the Hilbert-Huang transform (HHT) [Huang 2014]. Inspired by these works, Dong et al. [Dong et al. 2020] proposed a framework

to decompose motions into several motion primitives using a multivariate compatible method called the multivariate empirical mode decomposition (MEMD) [Rehman and Mandic 2010] and analyze these motion primitives with their biomechanical features in the instantaneous frequency domain. However, these methods for motion data decomposition cost high computational resources and require heavy manual post-processing by animators.

To solve these issues, we introduce a novel architecture for learning-based methods that help animators decompose and edit complex motion features in the instantaneous frequency domain with lower computational resources and high-quality outputs. Our proposed method enables animators to efficiently decompose motion into non-linear frequency motion components and edit these motion features in the desired frequency bands without manual post-processing. In summary, our study makes the following contributions:

- Architecture for the neural networks decomposing motion data in the instantaneous frequency domain based on MEMD with lower computational cost.
- Application of spectral motion editing for automatically synthesizing motion using decomposed non-linear motion components with their biomechanical features.

2 RELATED RESEARCH

Various motion decomposition studies in the frequency domain have been conducted, such as Yumer et al. [Yumer and Mitra 2016] proposed a method to edit motion features in the frequency domain. Since these methods only obtain decomposed data with mathematical meaning, the usages were restricted by their limitations, resulting in some primitive results.

HHT driven by MEMD [Rehman and Mandic 2010] is a non-linear decomposition method that is suitable for multivariate data in the frequency domain. To adopt HHT into motion editing and generation for computer graphics, Dong et al. [Dong et al. 2017] proposed a framework to decompose and edit dance motion using the non-linear processing property of MEMD-based HHT.

However, with the development of the deep learning method, researchers and animators need a custom-made neural network for MEMD-based HHT to fully employ its property by considering the non-linear and non-stationary characteristics of the complicated human data for motion synthesis. To challenge this problem, in this paper, we propose a novel motion learning network architecture to learn MEMD for non-linear motion decomposition with their biomechanical features, providing a novel editing technique in the instantaneous frequency domain.

3 MOTION REPRESENTATION

In this research, we define the motion data $M \in \mathbb{R}^{T \times 4J}$ as the input data to extract motion features for feeding the neural networks, where T indicates the time and J indicates the joint rotations represented by the unit quaternions. To make the non-linear motion data features can be preserved based on a hierarchical structure of skeleton joint rotations, which are human interpretable [Dong et al. 2020], we define the decomposed motion data $\tilde{M} \in \mathbb{R}^{T \times 3J \times N}$ using the Euler angles. Here, N indicates the number of IMFs decomposed by the neural network. Therefore, the input human motion capture

data M can be defined as (1) based on MEMD:

$$M = \sum_{i=1}^N \tilde{m}_i + R \quad (1)$$

where $\tilde{m}$ indicates a single decomposed IMF and R represents the residual (Trend) as the posture of the motion without periodicity. The IMF definition is shown below [Huang 2014]:

- The number of extremum and zero crossings in the data are equal or differ by one.
- At any given time, the average value of the envelope connecting the maximum value to the minimum value is zero.

Then, we can correctly obtain the instantaneous amplitude and frequency for each IMF, as shown in (2):

$$A = \sqrt{\tilde{M}^2 + \mathcal{H}[\tilde{M}]^2}, \Omega = \frac{d}{dt} \arctan \frac{\mathcal{H}[\tilde{M}]}{\tilde{M}} \quad (2)$$

where $\mathcal{H}[\]$ represents HT that assumes the real part of motion data are monochromatic waves $a(t) \cos(\omega(t)t)$, and converts the real part into its imaginary part [Bracewell and Bracewell 1986]. A indicates the instantaneous amplitudes, and Ω indicates the instantaneous angular velocities of each decomposed Euler angle in the complex plane.

4 LEARNING MOTION DECOMPOSITION

Figure 2 demonstrates our proposed architecture for learning motion decomposition in the instantaneous frequency domain. In the present research, we focus on three IMFs whose averaged frequency is around 5Hz, 2Hz, and 1Hz, respectively. These three frequency bands, 0.2s, 0.5s, and 1s motion, are the most important movements in human body movements.

Firstly, positional encoding (PE) proposed in NeRF [Mildenhall et al. 2021] is adopted to embed the input data into a frequency space more readily accessible, improving the neural network's ability to learn high-frequency features.

Secondly, high-frequency embedded motion data is fed into three motion decomposition encoders, $E_{\text{IMF}_{5\text{Hz}}}$, $E_{\text{IMF}_{2\text{Hz}}}$ and $E_{\text{IMF}_{1\text{Hz}}}$, respectively. Then, after passing through three latent spaces, three motion decomposition decoders $D_{\text{IMF}_{5\text{Hz}}}$, $D_{\text{IMF}_{2\text{Hz}}}$ and $D_{\text{IMF}_{1\text{Hz}}}$ output the non-linear decomposed motion features in the instantaneous frequency domain. This approach allows the network to learn the multivariate empirical mode decomposition at the latent spaces for each target decomposition frequency band.

Thirdly, we also employ the forward kinematics layers at the end of each decoder to let the losses be applied to joint positions and joint rotation used in the input data [Villegas et al. 2018]. Thus, the final output of each motion decomposition encoder is presented as $\text{IMF} \in \mathbb{R}^{T \times 3J}$ that included in the decomposed motion sets $\tilde{M} \in \mathbb{R}^{T \times 3J \times 3}$ defined in the above section. Then, the latent space for each encoder and decoder is represented by a feature matrix $Z \in \mathbb{R}^{T \times H}$ where H indicates the reduced joint dimension. Thus, the decomposed IMFs are obtained by learning the neural functions:

$$f_d(M) : \mathbb{R}^{T \times 4J} \rightarrow \mathbb{R}^{T \times 3J \times 3} \quad (3)$$

which gives input motion data with quaternion representation M . Thus, the non-linear motion decomposition based on MEMD was implicitly learned as $f_d(\cdot)$ by the neural networks.

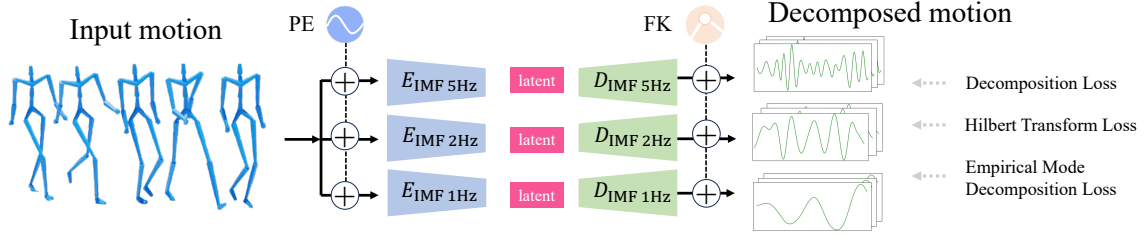


Figure 2: The proposed motion decomposition architecture based on MEMD and HT losses for spectral motion editing.

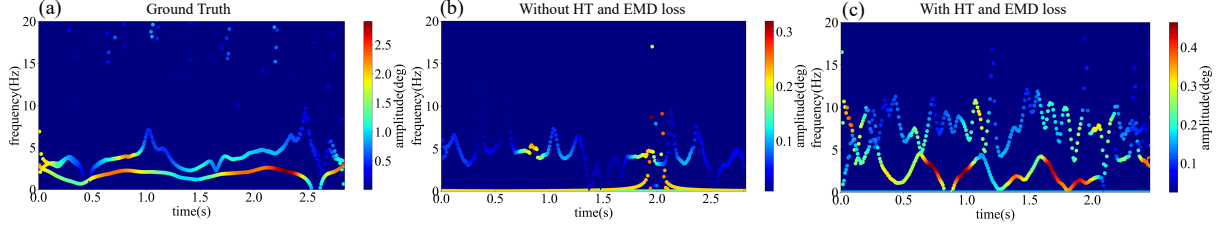


Figure 3: A left leg joint rotation decomposed by the proposed architecture. By comparing the two spectra obtained by the neural networks (b,c) with the ground truth spectrum (a), it can be seen that HT and EMD losses enable the network to better decompose motion features in the instantaneous frequency domain.

In this study, we adopt the one-dimensional convolutional neural network (1DCNN), which is commonly adopted in motion learning tasks with quaternion unit loss, twist loss, and reconstruction loss [Aberman et al. 2020; Holden et al. 2017]. However, novel loss functions must be presented to learn the multivariate empirical mode decomposition. First, a decomposition loss $\mathcal{L}_{dec}$ gives the most essential learning property.

$$\mathcal{L}_{dec} = \|\tilde{M} - \tilde{M}_{label}\|_1 \quad (4)$$

where $\|\cdot\|$ represents the L1 norm that allows the network to capture detailed motion features. Next, according to the definitions of IMF, we also propose an EMD decomposition loss $\mathcal{L}_{EMD}$, including a local mean loss $\mathcal{L}_{mean}$ and an extremum number $\mathcal{L}_{ext}$ loss.

$$\begin{aligned} \mathcal{L}_{EMD} &= \mathcal{L}_{mean} + \mathcal{L}_{ext} \\ &= \|\langle \tilde{M} \rangle_{ave} - \langle \tilde{M}_{label} \rangle_{ave}\|_1 + \|\text{extnum}(\tilde{M}) - \text{extnum}(\tilde{M}_{label})\|_1 \end{aligned} \quad (5)$$

where $\langle \tilde{M} \rangle_{ave}$ indicates a multivariate mean vector obtained by averaging the time for each joint angle to constrain the local mean to zero for each IMF. In addition, $\text{extnum}(\cdot)$ represents the number of local maximum and minimum, and $\text{zeronum}(\cdot)$ represents the number of zero crossings, respectively, making the decomposed IMFs have the same number of extremum with zero crossings.

Finally, we present an HT loss including an instantaneous frequency loss $\mathcal{L}_{freq}$ and an instantaneous amplitude $\mathcal{L}_{amp}$ using (2).

$$\mathcal{L}_{HT} = \mathcal{L}_{freq} + \mathcal{L}_{amp} = \|\Omega_{\tilde{M}} - \Omega_{\tilde{M}_{label}}\|_1 + \|A_{\tilde{M}} - A_{\tilde{M}_{label}}\|_1 \quad (6)$$

Therefore, our final loss $\mathcal{L}$ employed for training decomposition combines the above loss terms, as shown below:

$$\mathcal{L} = \mathcal{L}_{dec} + \alpha_{EMD} \mathcal{L}_{EMD} + \alpha_{HT} \mathcal{L}_{HT} \quad (7)$$

5 DATA PREPROCESSING AND LEARNING

Motion capture system records mark positions and then converts these motion data into Euler angles of each joint based on hierarchical skeleton structures. Then, in this study, we adopted a Hierarchical skeleton model with 30 joints, 90 degrees of freedom (DoF), and 120 frames as the sampling rate, available at the CMU Graphics Lab Motion Capture Database [CMU 2019] as the training data. Next, we decompose these motion data based on previous research [Dong et al. 2020] using MEMD. Finally, to prepare the motion data for the neural network, we divide motion data into a 240-frame window, which can capture greater than 1 Hz motion data according to the sampling rate of CMU. In addition, we also overlapped the motion data with 120-slider frames. Thus, after data preprocessing, we have both original M and decomposed IMFs data $\tilde{M}_{label}$ paired for learning.

Training is performed for 500 epochs and takes around 50 hours on an NVIDIA RTX A4000 / 16GB GPU. The learning rate was set to 1×10^{-4} with the Adam optimizer, and $\alpha_{EMD} = 2$ and $\alpha_{HT} = 2$ were adopted in our experiments.

6 RESULTS

In this present study, we employed gait motions as an example to demonstrate our proposed architecture for learning MEMD. Figure 3 shows an example of a joint rotation obtained by our neural network. In a MEMD-based motion decomposition, it is necessary to extract different non-linear biomechanical motions in different frequency bands. That is, distinct IMFs were shown separately in the ground-truth spectrum (a). However, it is difficult to learn the motion features that were embedded into their different non-linear frequency bands without our proposed EMD and HT losses since the amplitudes were not evenly distributed roughly, and frequencies

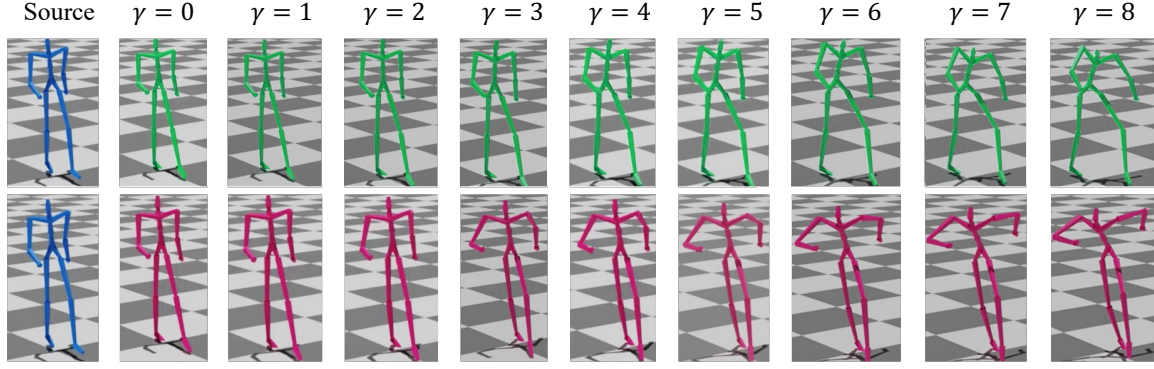


Figure 4: An example of spectral motion editing (a gait motion where $T = 1$). By separately editing the amplitudes of decomposed IMFs decomposed from the source motion (blue) by the neural network, two different frequency bands of IMF 2Hz (green) and IMF 1Hz (red) corresponding to their biomechanical features were exaggerated, respectively.

were mixed together (b). After applying our proposed losses, the neural network can better learn the process of decomposing the most important IMFs under 10 Hz (c).

Next, our proposed architecture could be employed for spectral motion editing by combining our method with other motion editing networks. For example, by introducing our method into a style-transfer neural network [Aberman et al. 2020] to train an editing network function $f_e(M, f_d(M)) : \mathbb{R}^{T \times 4J} \rightarrow \mathbb{R}^{T \times 3J}$. Figure 4 shows a spectral editing example using the combined motion editing network. For simplicity, we only show two IMFs (2Hz and 1Hz) editing results since IMF 5Hz, 0.2s motions, contains less important biomechanical features during the walking movements. As can be seen from the results, by introducing a γ as a hyperparameter to adjust the amplitude of the corresponding frequency bands by $f_e(M, \gamma f_d(M))$, we could achieve a high-quality character animation with non-linear frequency edited by the neural network.

For instance, the IMF 1Hz represents the 1-second shoulder and leg motions regarding the walking steps, and the IMF 2Hz represents the 0.5-second arm and ankle motions regarding the secondary motion. Then, the results showed that after enhancing the amplitude of different IMFs with γ from 0 (reconstruction) to 8, edited motions with other tastes corresponding to the distinct walking rhythm could be achieved. Thus, since other types of human motion could also be decomposed into several IMFs that are motion primitives containing biomechanical features, our method could be adapted to other motion data for spectral motion editing. Furthermore, it usually takes 20 seconds to decompose data directly using MEMD using a personal computer, which causes high computational resources. In comparison, only about 2 seconds were generally required by the neural network editing, including post-processing. Since users need to interactively edit γ to obtain the desired motion, the high-performance decomposition we achieved is important for usability.

7 CONCLUSION

Learning and editing non-linear motion features in the frequency domain is challenging for synthesizing character animation. Our

proposed neural network architecture could decompose the important frequency bands with their biomechanical features, achieving spectral motion editing in the frequency domain. The learning-based approach also provides a more interactive editing experience. Our application results revealed that our architecture could benefit many applications for character animation design.

The limitation of the present work is a lack of quantitative evaluations for motion decomposition and motion editing in the instantaneous frequency domain. Further experiments and quantitative indicators are required in future work.

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